



Terrestrial ecosystem recovery following removal of a PCB point source at a former pole vault line radar station in Northern Labrador

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HIGHLIGHTS

- Arctic ecosystem recovery following removal of PCB point sources.
- PCB concentrations in birch, willow and deer mice tissues are tracked over 14 years.
- PCB levels in birch and willow samples are ca. 4x lower in 2011 than 1997.
- PCB levels in deer mice are ca. 3x lower in 2011 than in 1997/98.
- Residual PCB levels are not likely to decrease in the foreseeable future.

ARTICLE INFO

Article history:

Received 18 March 2013

Received in revised form 22 April 2013

Accepted 23 April 2013

Available online 25 May 2013

Editor: Frank Riget

Keywords:

Ecosystem recovery

Long-term monitoring

Polychlorinated biphenyls

Birch

Willow

Deer mice

ABSTRACT

Saglek Bay (LAB-2), located on the northeast coast of Labrador is a former Polevault station that was operated by the U.S. Air Force from 1953 to 1971 when it was abandoned. An environmental assessment carried out in 1996 determined that the site was contaminated with polychlorinated biphenyls (PCBs) with concentrations in soils far exceeding the Canadian Environmental Protection Agency (CEPA) regulation of 50 µg/g in three areas of the site (Beach, Site Summit, Antenna Hill). This led to remediation work carried out between 1999 and 2004 to remove and/or isolate all PCB-contaminated soil exceeding 50 µg/g and to further remediate parts of the site to <5 µg/g PCBs. In this study, spatial and temporal trends of PCB concentrations in soil, vegetation (*Betula glandulosa* and *Salix* spp.), and deer mice (*Peromyscus maniculatus*) were investigated over a period of fourteen (1997–2011) years in an effort to track ecosystem recovery following the removal of the PCB point sources. The data collected shows that PCB levels in vegetation samples are approximately four times lower in 2011 than pre-remediation in 1997. Similarly, PCB concentrations in deer mice in 2011 are approximately three times lower than those measured in 1997/98. Spatial trends in vegetation and deer mice continue to demonstrate that areas close to the former point sources of PCBs have higher PCB concentrations than those further away (and higher than background levels) and these residual PCB levels are not likely to decrease in the foreseeable future given the persistent nature of PCBs in general in the environment, and in particular in cold climates.

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1. Introduction

Saglek Bay, Labrador is the site of a former Polevault Line radar station dating from the 1950s, and a current long-range radar facility that was built as part of the Canadian North Warning System in the mid-1980s. The site (known as LAB-2), is located on the northeast coast of Labrador in the Torngat Mountains (58°23'N, 62°35'W), approximately 600 km from Goose Bay (Fig. 1). The former Polevault station was operated by the U.S. Air Force from 1953 to 1971 when it was abandoned and subsequently demolished. In the mid-1980s a

new radar facility was constructed at the same location. During an environmental assessment of the site in 1996, it was determined that three areas of the site (Beach, Site Summit, Antenna Hill) were contaminated with PCBs at concentrations in excess of 50 µg/g. Local weather and wind patterns affected contaminant re-distribution in the area, creating a 'halo of contamination' up to 27 km away from LAB-2 (Bright et al., 1995; Dushenko et al., 1996; Pier et al., 2002).

Soils, plants (i.e. birch and willow), lake sediments, and local wildlife (i.e. deer mice) collected in 1997/98 prior to site remediation showed a negative log–linear relationship of PCB concentration with increasing distance from the contaminant source at LAB-2, and a distinct Aroclor 1260 congener signature pattern (Brown et al., 2009; Johnson et al., 2009; Pier et al., 2003). Aroclor 1260 was the primary commercial PCB mixture used at Polevault sites on the Canadian East Coast whereas Aroclor 1254 (which has a higher proportion of lighter congeners

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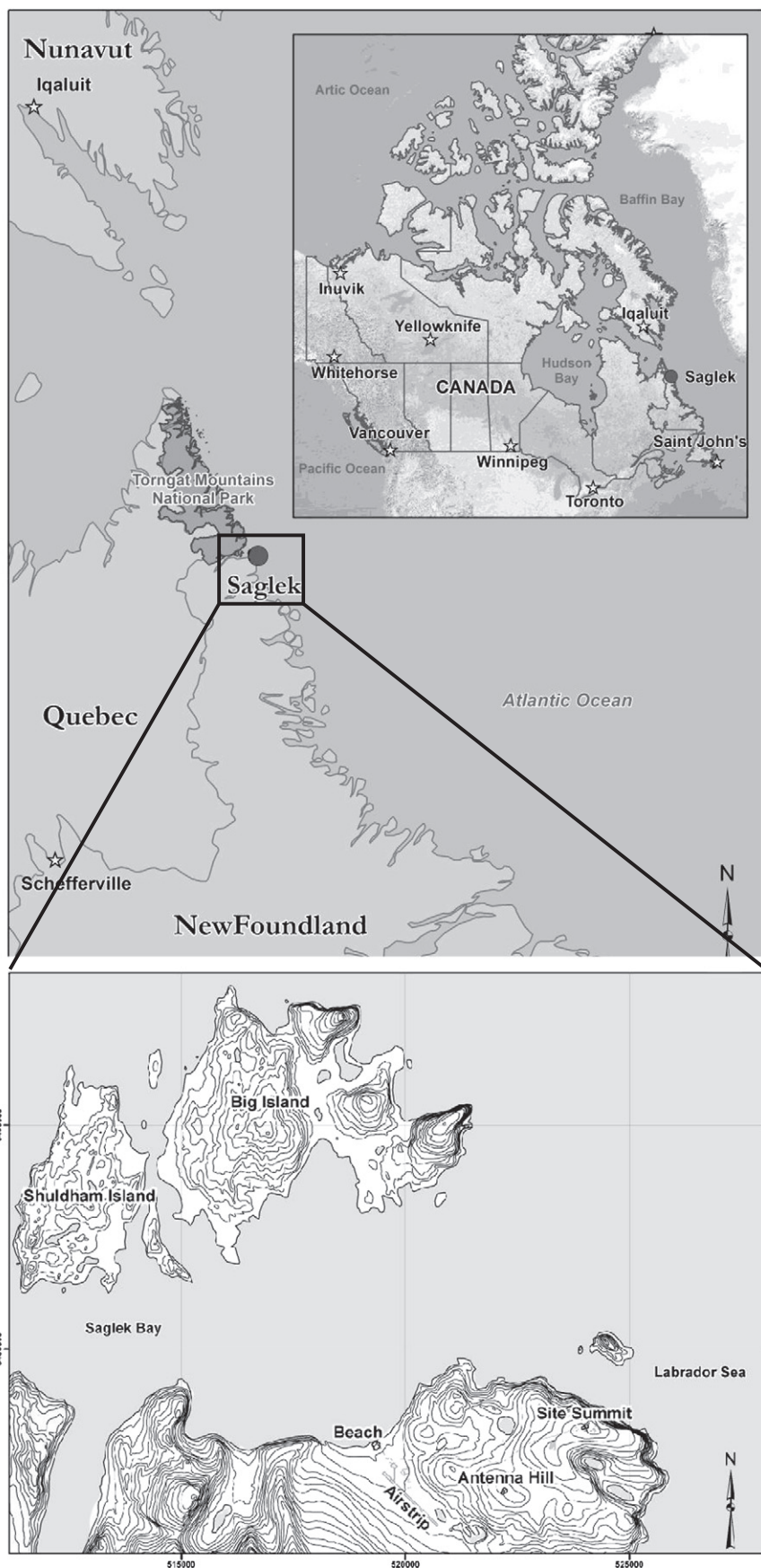


Fig. 1. Location of three former PCB point sources (Beach, Antenna Hill, and Site Summit) at Saglék, Labrador (LAB-2) and background sites (Big Island and Shuldham Island).

compared to 1260) was used at other radar sites in the Canadian Arctic (Bright et al., 1995; Dushenko et al., 1996). Thus, PCB congener signature patterns can be used to determine if contamination is from short-range (Aroclor 1260), long-range from other arctic sources (Aroclor 1254) or from southern latitudes (Bright et al., 1995; Dushenko et al., 1996; Choi et al., 2008).

Remediation work was conducted between 1999 and 2004 to excavate, stockpile (at a 'staging area'), and then ship off-site all PCB-contaminated soil ($>50 \mu\text{g/g}$) from the Beach, Site Summit and Antenna Hill. The Site Summit and Antenna Hill were subsequently capped with clean soil, while soil at the Beach was remediated to $<5 \mu\text{g/g}$ due to proximity to the sensitive marine environment (Brown et al., 2009). After completion of remediation work at LAB-2 in 2004, a long-term monitoring (LTM) program involving the collection and analyses of vegetation and deer mice was designed and implemented to monitor ecosystem recovery. Plants have previously been shown to accumulate PCBs in shoot and root tissues in both temperate and arctic environments (Dushenko et al., 1996; Ficko et al., 2010; Pier et al., 2003; Whitfield-Åslund et al., 2007; Zeeb et al., 2006) and thus were chosen to provide a measure of current contaminant concentrations at the primary level of the terrestrial food chain. Plants also act as natural passive air samplers and generally reflect current aerial concentrations of contaminants in comparison to soils which are more reflective of long term PCB deposition and accumulation over time (Luttmer et al., 2013; Pier et al., 2003). As LAB-2 is located in a tundra zone above the treeline, dwarf shrubs, mosses and lichens are the predominant vegetation in established areas, whereas early succession species can be observed in disturbed areas.

Dwarf birch (*Betula glandulosa*) leaves were chosen to monitor the temporal changes in PCB concentrations on a yearly basis (Luttmer et al., 2013) as they are a deciduous species and widespread across the site. Analysis of PCB concentrations in leaves (new shoot growth) collected at the end of the growing season represents the net PCB accumulation on a yearly basis, and may indicate if aerial PCB concentrations are decreasing over time. Willow plants (*Salix* spp.) were selected to compare temporal changes in PCB concentrations and congener patterns in woody perennial plants from 2011 to those collected in 1997/98, prior to remediation (Pier et al., 2003). In 2011, roseroot (*Rhodiola rosea*) and mountain sorrel (*Oxyria digyna*), two edible herbaceous species were selected to investigate the current spatial trends in PCB concentrations between plants growing directly in and around the excavated areas of the Beach, to plants from further away. Comparison of PCB concentrations in shoot tissues of these early succession species found growing on the disturbed areas following remediation will indicate the extent of residual contamination remaining in the soil. Finally, deer mice (*Peromyscus maniculatus*) were selected to monitor temporal and spatial trends in PCB concentrations at the primary consumer level. In alpine and arctic environments, deer mice are found in microhabitats consisting of moderate amounts of rock and vegetation cover on slopes that are generally less than 30° , and with home ranges of ca. 1280 m^2 (Bowers and Smith, 1979; Reichel, 1986). Higher concentrations of PCBs are often found in fine soil material that collects in the lee of rocks and boulders, which is also typical habitat for mice; therefore regional differences in PCB concentrations in the mice around Saglek are expected based on the distribution of historic and residual PCB contamination. Deer mice collected as part of the LTM program were living in the areas with the highest residual contamination, and are therefore good indicators of the worst-case (highest) residual PCB contamination on-site.

At LAB-2, the concentrations of other organic and inorganic contaminants are low and therefore the site provides a unique opportunity to investigate the recovery of the terrestrial ecosystem from chronic exposure to a single contaminant (ESG, 1997, 1999; Johnson, 2010). However, the time-frame required for natural attenuation of PCBs in a terrestrial Arctic ecosystem to background levels following removal of a PCB point source has yet to be determined. According to Smol (2008), the majority of

published environmental monitoring studies are based on observations from one year or less, though long-term studies of six or more years are often necessary to put sampling results in context of natural environmental variability. Thus this study is one of the first to investigate trends in both vegetation and small mammals to assess long-term ecosystem recovery following PCB-contamination in the Arctic.

2. Materials and methods

2.1. Sample collection

Dwarf birch (*B. glandulosa*) and willow (*Salix* spp.) samples were collected in July and August in 1997 and 1998 from multiple transects extending from 50 m up to 1.2 km away from each of the contaminated areas with PCB concentrations above $50 \mu\text{g/g}$ (Beach, Antenna Hill and Site Summit), as well as from reference locations up to 7 km away from the site (Fig. 1) (Pier et al., 2003). Samples were comprised of a composite of roots, shoots and leaves ($\sim 20 \text{ g}$ total wet weight) from a single plant collected within 1-m^2 of a corresponding soil sample. Plants were washed under running water on-site, patted dry, then wrapped in aluminum foil, placed in Ziploc™ bags and kept frozen until analysis. Soils were collected from the top 10 cm of substrate using a prewashed stainless steel scoop and placed in a 125-mL amber jar (Proclean™) fitted with a Teflon-lined lid.

Birch leaf monitoring began in 2001 north of the staging area and in 2006 southeast of the Beach, and was conducted annually or bi-annually until 2010. Each collection year, ten samples composed of approximately 20 g of unwashed birch leaves were collected in mid or late August from within a 15 m radius of the monitoring location, wrapped in aluminum foil and placed in labeled Whirlpak bags and kept frozen until analysis. Samples were also collected from background locations on Big Island and Shuldham Island, located 6–7 km northwest of the site (Fig. 1). Birch leaf samples were not collected at the Site Summit or Antenna Hill because these areas are predominantly exposed bedrock and boulders and there is insufficient vegetation for annual monitoring.

In August 2011, willow samples were harvested at varying distances from 50 m up to 1 km away from the formerly contaminated areas (Fig. 1), and at background locations. A remote background sample was also collected inland of the site in the Tornat Mountains National Park ($\sim 40 \text{ km}$ northwest of LAB-2). This location was selected as it is separated from the site by a mountain chain, and thus was not expected to be within the area of influence ("halo") of the site (Pier et al., 2003). Samples collected from each location in 2011 consisted of a soil sample, root tissue and shoot tissue. Plants were separated into root, shoot and leaf tissues using scissors, which were rinsed with methanol between cuts. All tissues were washed under running water on-site, air-dried, weighed, and then wrapped in aluminum foil, placed in labeled Whirl-Pak® bags and kept frozen until analysis.

Roseroot (*R. rosea*) and mountain sorrel (*O. digyna*) samples were also harvested from around the site and from site background and remote background locations in August 2011. Roseroot samples were collected in and directly adjacent to the excavated area at the Beach, while mountain sorrel samples, which was more widespread around the site, were collected in and directly adjacent to the excavated area at the Beach, as well as from the staging area, around the Site Summit cap, southeast (350–550 m) of the Site Summit cap, from a site reference location at the south end of the Airstrip, from the site background locations and from the remote background location. Shoot samples were harvested using scissors that were rinsed with methanol between samples. Samples were shaken lightly to remove any soil particles that were adhered to the plant, but were not washed prior to analysis. Samples were then wrapped in aluminum foil, weighed, placed in labeled Whirl-Pak® bags and kept frozen until analysis. Soil samples from 2011 were collected and stored as in 1997/98.

Deer mice (*P. maniculatus*) were collected at the Site Summit, Antenna Hill, east and west of the Beach and from north of the Staging Area as part of the LTM (Fig. 1). All necessary research and collection permits were obtained and all conditions of the permits were followed, including handling requirements. The Beach, Site Summit and Antenna Hill were selected because these areas had the highest levels of historic PCB contamination. Between 2002 and 2005, during the temporary storage and remediation of the stockpiled contaminated soils, mice were monitored north and south of the staging area as part of the Environmental Effects Monitoring Program (ESG, 2005). Long-term monitoring was continued in 2006 for the location north of staging area and incorporated into the LTM. In some years, mouse samples were also collected near the water supply lake, at the south end of the airstrip, and on Big Island (Fig. 1). These areas are considered site references, but because they are within the area of historic aerial influence (Pier et al., 2003) PCB levels may be elevated compared to remote areas. Samples were also collected west of the Beach in 2006 and 2007 to examine potential differences in PCB concentrations on either side of the excavated ravine at the Beach (ESG, 2008).

Mice were trapped mid-August each year using snap traps baited with peanut butter and raisins. Traps were placed within a 15 m radius of the designated monitoring site location. Traps were set in the evening and checked each morning for a week or until 10 samples were collected. Mouse samples were given independent sample numbers and morphological measurements were recorded prior to freezing the samples until analysis.

2.2. Soil and tissue analysis

2.2.1. Congener analysis

All willow samples, whole mouse samples and corresponding soil samples were analyzed for 91 PCB congeners and percent lipid by AXYS Analytical Services Ltd., Sidney, BC. Mouse samples collected prior to 2005 were composited, while livers from mice collected in 1997 were analyzed separately from the rest of the mouse tissue as PCBs are preferentially concentrated in fatty tissues such as the liver. PCB congener concentrations were determined by high-resolution gas chromatography with detection by high- (1997/98) and low- (2011) resolution mass spectrometry (HRGC/LRMS) using in-house method MLA-007 (described in detail in Pier et al., 2003). Congener detection limits were sample-specific and ranged from 0.01 to 1 ng/g. All sample concentrations were reported as ng/g wet weight. As analytical techniques improved between 1997/98 and 2011, two sets of co-eluting congeners (28/31 and 84/92) from 1997/98 were reported as individual congeners in 2011. When comparing data between years, the 2011 congeners were combined to form a single concentration as described in Pier et al. (2002).

2.2.2. Total Aroclor analysis

Birch leaves, roseroot and mountain sorrel samples (and corresponding soils) were analyzed for total PCBs (Aroclor 1260 equivalents, dry weight) and percent lipid by the Analytical Services Unit (ASU) at Queen's University, Kingston, ON using IASU Method 01 described in Zeeb et al. (2006) and Whitfield-Åslund et al. (2007). Two samples from the Beach (one roseroot and one mountain sorrel) were divided in half for comparative analysis and sent to both ASU for analysis by Aroclor 1260 equivalents and to AXYS for analysis of PBC congeners (wet weight) by high resolution gas chromatography with detection by high-resolution mass spectrometry (HRGC/HRMS) (data unpublished). Soil samples analyzed at ASU were homogenized then subsampled to determine a wet/dry ratio, while plants were finely chopped and dried overnight in a vented oven at 25°C. All samples were then extracted by soxhlet extraction for 4–6 h and analyzed for total PCBs using an Agilent 6890 Plus gas chromatograph with a ^{63}Ni electron capture detector (GC/ECD) with a detection limit of 5 ng/g. Percent lipid was determined by gravimetric lipid analysis. Percent moisture was determined for all

samples analyzed at ASU so that dry weight concentrations could be converted to wet weight for comparison between analytical methods. Results of this and other samples previous analyzed by both labs (data not published) showed minimal differences in concentrations.

2.3. Data analysis

2.3.1. Birch samples

Statistical analyses for birch samples were conducted using SPSS Software Package (Version 20.0, SPSS Inc., 2011). The data was tested for normality and transformed as appropriate for parametric analyses. Plant samples were not lipid-normalized as in previous studies (Brown et al., 2009; Pier et al., 2002) as the percent lipid did not correlate with the log-transformed PCB concentration. Therefore, all birch samples were reported on a dry-weight basis. A significance level of $\alpha = 0.05$ was used for all tests, results were recorded with the arithmetic mean and standard error, and PCB concentrations are reported to three significant figures.

2.3.2. Willow, mountain sorrel and roseroot samples

Willow, mountain sorrel and roseroot samples were analyzed identically to birch samples except statistical analyses were conducted using Systat Statistical Software Package (Version 13.0, SPSS Inc., 2009). As detailed in Pier et al. (2003), samples were subdivided into groups based on their proximity to the local PCB source at Saglek (on-site: <3 km, site background: 6–10 km off-site, and remote Arctic background: >25 km off-site).

Total PCB concentrations in willows were determined based on the sum of the congeners that were detected in 50% or more of the samples. This resulted in 53 congeners (38 peaks) for the 2011 data and 57 congeners (39 peaks) for the 1997/98 data set. In general, the two data sets were quite similar. Once the data sets were reduced to 53/57 congeners, those congeners with concentrations less than the detection limit were replaced with a random number between zero and the detection limit.

Prior to comparison between the two willow data sets, the individual root, shoot, and leaf concentrations for each sample in the 2011 data set were converted to an average whole plant concentration for each sample. The total plant concentration was determined as the sum of congeners based on the proportional weights of the root, shoot, and leaf samples using the following equation:

$$\sum_{i=0}^{57} \text{congener} = \left(\frac{\text{mass of root}}{\text{total mass}} \times [\text{PCB}]_{\text{root}} \right) + \left(\frac{\text{mass of shoot}}{\text{total mass}} \times [\text{PCB}]_{\text{shoot}} \right) + \left(\frac{\text{mass of leaf}}{\text{total mass}} \times [\text{PCB}]_{\text{leaf}} \right).$$

2.3.3. Mice

Statistical analyses for all mice samples were conducted using SPSS Software Package (Version 13.0, SPSS Inc., 2011). Analysis was identical to the birch samples, except mice samples were lipid-normalized as the percent lipid correlated with the log-transformed PCB concentration. In order to make comparisons between the pre-cleanup (1997) and post-cleanup conditions, whole body PCB concentrations for samples collected in 1997 were determined based on a weighted average of the composited liver and body samples.

3. Results and discussion

3.1. Dwarf birch

In 1997, the mean concentration of PCBs in *B. glandulosa* tissues was 18 ng/g (wet weight). As these plant samples were collected between 150 and 300 m from the Beach excavation they are, as expected,

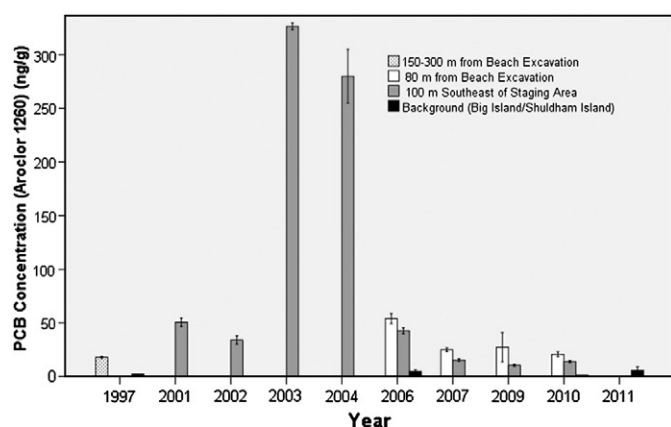


Fig. 2. PCB concentrations in birch samples from 1997 to 2011. The samples collected in 1997 were whole plant samples and concentrations are based on wet weight. All other samples are new-year growth and concentrations are based on dry weight. Error bars represent the standard error of the mean.

somewhat lower than those samples collected in subsequent years on or closer to the direct source of PCBs at the Beach. PCB concentrations in the birch leaves at the Beach decreased from a mean of 54 ng/g in 2006 to 21 ng/g in 2010 while concentrations at the Staging Area decreased from a mean of 51 ng/g in 2001 to 14 ng/g in 2010 (Fig. 2). Significant increases in PCB concentrations in the birch leaves to levels greater than 280 ng/g were observed at the Staging Area in 2003 and 2004. These increases were associated with aerial redistribution of PCBs during the screening of contaminated soils at the staging area. Levels decreased significantly to 40 ng/g in the following monitoring period (2006), demonstrating the sensitivity of the vegetation as a passive air sampler and that the leaves are a good measure of aerial deposition within a growing season (Luttmer et al., 2013).

Previous studies have indicated that a “halo” of contamination exists up to 25 km away from the site, and is influenced by predominant wind and storm patterns (Pier et al., 2003). In 2006, the mean PCB concentration in birch leaves at Shuldham Island (background reference location) was 5.0 ng/g, while in 2010 all three samples were below detection limits (<0.30 ng/g). The decrease in PCB concentrations observed on Shuldham Island indicates that aerial distribution of PCBs to areas further away from the site has decreased. However, in 2011, the mean concentration in birch leaves collected from Big Island (second background location) was 5.8 ng/g. It is likely that Big Island receives a greater PCB input from aerial distribution of PCBs than Shuldham Island, due to the direction of prevailing winds in the area. It should be noted however, that all of these concentrations are very low in general and specifically compared to plants growing <1 km from a former point source.

Since 2007, PCB concentrations in the birch leaves at the Beach and staging area monitoring locations have not shown any significant changes and are beginning to stabilize at about 10–20 ng/g (Fig. 2). Although the concentrations on-site are elevated compared to background samples (<3 ng/g; 5 ng/g) it is unlikely that they will decrease to background levels due to low-level residual PCB contamination in the soil on the site. The PCB levels observed in the birch leaves are likely a result of uptake of PCBs from the soil via the roots and translocation within the plant (Luttmer et al., 2013). The stabilization of PCB levels in leaves over the past few years to very low levels indicates that removal of the PCB point sources at the site has had the desired effect of limiting aerial distribution around the site.

3.2. Willow concentrations

PCB concentrations in willow plants were based on analysis of the whole plant (root and shoot tissues), and therefore higher concentrations than in the birch leaves are expected as sample includes tissues that are in direct contact with the soil (i.e. roots) as well as those affected

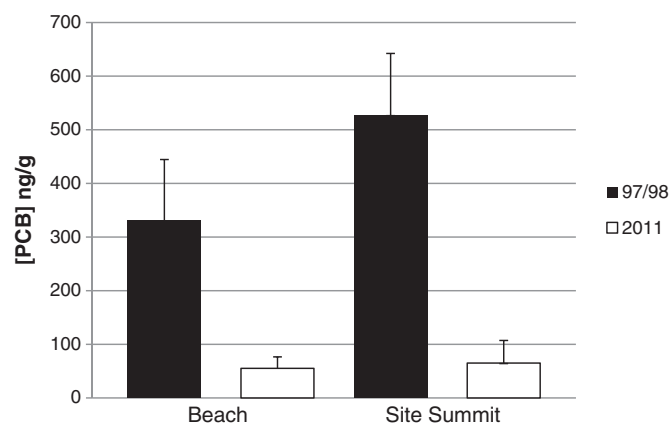


Fig. 3. Comparison of mean PCB concentrations in willows from the Beach and Site Summit in 1997/98 and 2011. Error bars represent the standard error of the mean.

by aerial deposition and/or uptake and translocation of PCBs (i.e. shoots). Following remediation, the whole tissue concentrations for willow plants collected within 200 m of the Beach and Site Summit in 2011 were six and eight times lower respectively than samples collected in 1997/98 (Fig. 3, Supplementary Information Table A).

In 1997/98, a characteristic Aroclor 1260 congener signature pattern was observed in soils and plants around LAB-2 (Fig. 4) and up to 27 km away (Bright et al., 1995; Pier et al., 2003), indicating that PCBs were being transported by air from the site and contaminating the environment by short range atmospheric transport. Following remediation of the PCB point sources, it was expected that aerial distribution of PCBs would decrease over time, which was seen as a decrease in the total PCB concentration of whole willow samples between 1997/98 and 2011 and in birch leaves during the same period. The similarity between the Aroclor 1260 congener profile and the on-site congener profiles in plants from the two sampling periods (Fig. 4) indicates that the plants continue to be affected by the former local source of contamination. Residual PCB contamination in the soils at Saglek is not expected to change, as PCBs typically degrade very slowly in Arctic environments (Meijer et al., 2003). Plants typically reflect the soil they are growing in (Aken et al., 2010), and hence it is anticipated that PCB concentrations in plants at Saglek will remain at current levels for a long time.

3.3. Roseroot and mountain sorrel concentrations

Mean PCB concentrations in roseroot ($n = 5$) and mountain sorrel ($n = 9$) samples collected in and directly adjacent to the excavated areas at the Beach were 320 ± 490 ng/g and 330 ± 310 ng/g, respectively (Supplementary Information Table B), with a maximum concentration of 1190 ng/g detected in roseroot, and 730 ng/g in mountain sorrel. While these concentrations were higher than those observed in willow samples, these plants were collected directly from and adjacent to the excavated areas at the Beach which likely has higher residual concentrations than vegetation from the surrounding area due to rocky outcrops on the cliff face where it was not feasible to excavate small pockets of contaminated soil. Variation in tissue concentrations is not unexpected as plant uptake of contaminants such as PCBs appears to be an inadvertent process associated with nutrient acquisition, which is also influenced by local environmental factors such as plant age, soil properties, temperature, soil water content and light regimes. A study by Ficko et al. (2010) documented large variations in PCB concentrations in shoot tissues of multiple weed species in southern Ontario likely due to environmental variation. Similarly, environmental factors and metal concentration in soil are known to influence plant uptake of metals into plant shoot tissues (Boyd et al., 1999; Jung, 2008; Sharma et al., 2007).

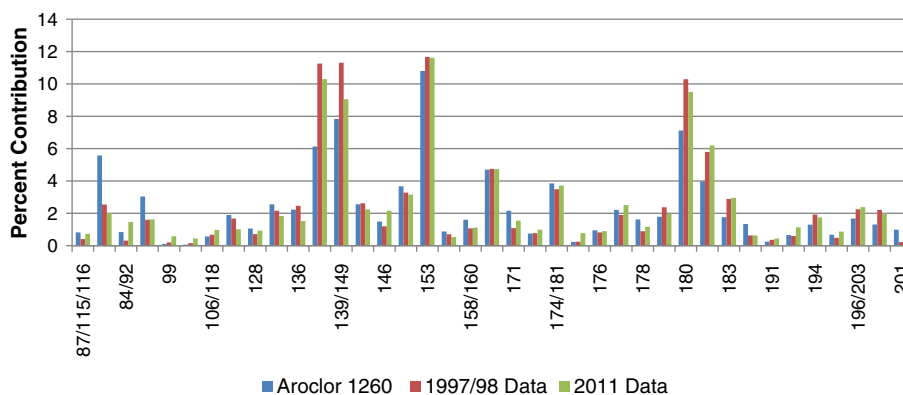


Fig. 4. Comparison of the Aroclor 1260 congener profile to the congener pattern in plants collected on-site in 1997/98 and 2011. Mean soil concentrations from the root zone of all willow samples were 640 ± 1000 ng/g in 1997/98 and 630 ± 1200 ng/g in 2011 ($n = 14$ and 10 respectively). Mean plant concentrations were 190 ± 250 ng/g in 1997/98 and 31 ± 38 ng/g in 2011.

Comparison of spatial trends on- and off-site determined that the mean concentrations in mountain sorrel were highest at the Beach (330 ± 310 ng/g), followed by the staging area (50 ± 32 ng/g), Site Summit (39 ± 22 ng/g), south end of the Airstrip (9.7 ± 9.7 ng/g), and site background samples from Shuldham and Big Islands (9.5 ± 9.8 ng/g). Similar to the willow samples, no PCBs were detected in the remote Arctic background sample. These results corroborate with the birch and willow results to demonstrate that even edible plants from areas close to, and on the former point sources, continue to show the effects of residual soil contamination to a greater degree than plants from further away. Current PCB levels in on-site vegetation are expected to persist for the foreseeable future as the PCB contamination is likely from root uptake of residual PCBs in the soil.

3.4. Mice concentrations

Capture rates for deer mice during the LTM program (2006–2011) were variable (Supplementary Information Table C). The lowest capture rates occurred in 2009, which is in agreement with wildlife surveys that indicated populations of small mammals were at a low throughout Labrador in 2009 (Philips, 2009). Differences in capture rates between

locations within the same year indicate that other factors were controlling mice populations at different locations around the site. Low capture rates in all years at the Site Summit were corroborated by field observations of very little suitable deer mouse habitat in this area (Supplementary Information Table D).

PCB concentrations in deer mice indicate that there is large variability within individual sampling events (e.g. same year and location) (Fig. 5; Supplementary Information Tables C and D). As mice have a small home range and primarily feed on vegetation, large variations are expected, based on the variability in both soil and vegetation concentrations at LAB-2. Age and feeding habits could also influence PCB concentrations. The large variability in concentrations in mice confirms the necessity of large sample sizes to interpret results and make inferences about site conditions.

Temporal trends in PCB concentrations in mice from east of the Beach have shown a decrease over time, from a mean of 17,000 ng/g in 1997 to approximately 7000 ng/g between 2006 and 2008 and down to 1400 ng/g in 2010 (Fig. 5; Supplementary Information Table D). In 2011, the mean concentration at the Beach was 4600 ng/g, but this was based on a much smaller sample size. PCB concentrations in Beach mice during all monitoring years between 2004 and 2011 were

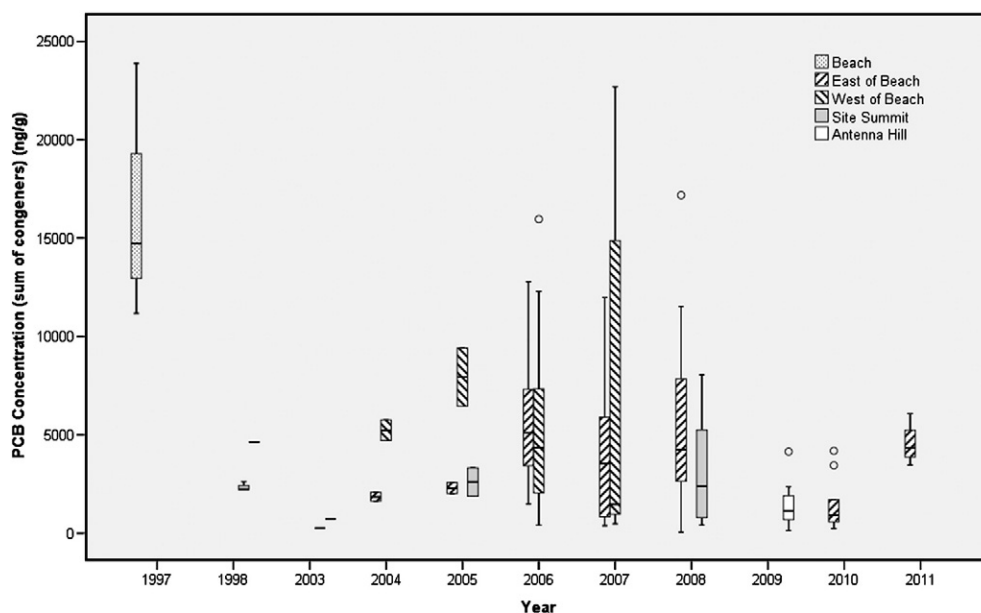


Fig. 5. PCB concentrations in mice collected on- and off-site between 1997 and 2011. Livers from mouse samples collected in 1997 were analyzed separately. Mouse samples collected before 2005 are composites.

significantly lower than the pre-remediation concentration in 1997. Temporal trends could not be evaluated at the other locations on-site due to insufficient capture rates over several years. Furthermore, hazard quotients (HQ) have decreased from 3.2 in 1997/98 to 2.1 in 2006; despite this decrease, there were associated biological effects, namely decreased bone mineral density (Johnson et al., 2009). The current (2010) HQ is 0.4, suggesting the risk has significantly diminished (Johnson, 2010). To investigate spatial trends, mice concentrations were compared among areas near the former point sources (Beach, Site Summit, and Antenna Hill) and other areas around the site (north of the staging area and south end of the airstrip; Supplementary Information Table D). Mean concentrations of mice collected near the former point sources were highest with concentrations of 7900, 4000, 3400, and 1400 ng/g, for Beach east, Beach west, Site Summit and Antenna Hill, respectively, while the mice samples collected at the staging area were lower (830 ng/g) and concentrations were even lower (77 ng/g) at the south end of the Airstrip (located ~1 km from a point source). Background samples collected on Big Island in 1997 and 2011 both had mean concentrations of approximately 30 ng/g. Spatial trends in mice resemble trends in vegetation, whereby samples collected near a point source are several orders of magnitude higher than samples collected further away.

To date, little research has been conducted on ecosystem recovery following removal of a contaminant point source. In this study, spatial and temporal trends in vegetation and mice were investigated over fourteen (1997–2011) years. Despite PCB concentrations in soil being approximately four times lower in 2011 than the pre-remediation levels from 1997/98, all vegetation sampled on-site in 2011 (birch leaves, willows, roseroot and mountain sorrel plants) continued to have low-level PCB concentrations. Similarly, PCB concentrations in mice in 2011 were approximately three times lower than those reported in 1997/98, and an HQ < 1 suggests minimal risk. While spatial trends in roseroot, mountain sorrel and mice continue to demonstrate that areas closer to former sources have higher concentrations than areas further away, temporal trends in birch leaves, willows and mice all indicate a decrease in concentrations over time. Based on the persistence of PCBs in general in this northern environment (over 40 years) and these long-term monitoring results (14 years), it is unlikely that the PCB concentrations at LAB-2 will decrease to the remote background concentrations in the foreseeable future.

Supplementary data to this article can be found online at <http://dx.doi.org/10.1016/j.scitotenv.2013.04.075>.

Acknowledgments

The authors are indebted to members of the Environmental Sciences Group (ESG) who have been involved in the Saglek project over the years. We are grateful for the help provided in the field by local wildlife monitors and contractors at the North Warning Systems Office (NWSO) LAB-2 site, and the funding provided by NWSO for this project. In addition, we would like to thank the Labrador Inuit Association (now the Nunatsiavut Government), Environment Canada and the Government of Newfoundland and Labrador, along with Canadian Wildlife Service, Geological Survey of Canada – Atlantic Region, Department of Fisheries and Oceans, Inuit from communities in Labrador, the captain and crews of long-liner vessels from Nain, the LAB-2 site contractor (Nasittuq) and

various other contractors for their input towards the successful completion of this project.

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